UNIT I: ENVIRONMENT AND ECOSYSTEMS

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MULTIDISCIPLINARY NATURE OF ENVIRONMENTAL STUDIES

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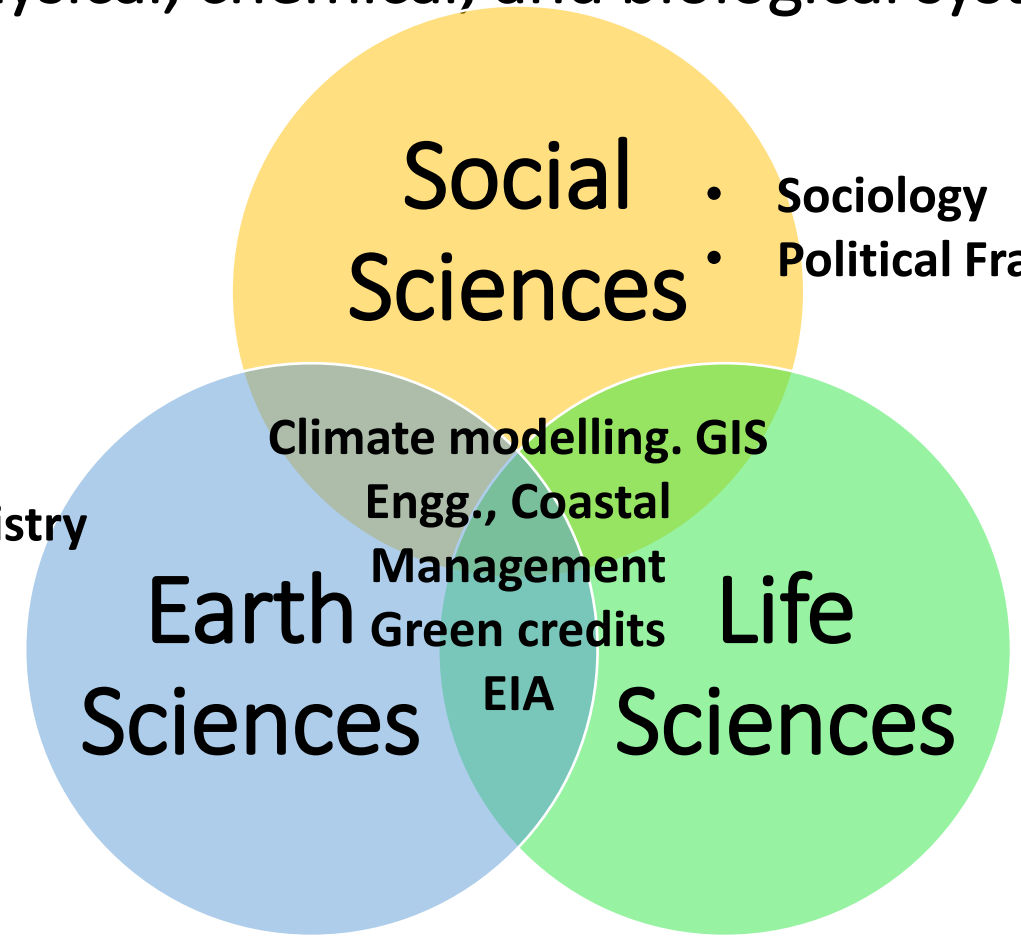
- To encircle/ surround
- interactions between physical, chemical, and biological systems



- Sociology
- Political Framework

- Meteorology
- Atmospheric Chemistry
- Hydrology
- Geology
- Geomorphology

- Soil Sciences
- Ecology
- Population dynamics
- Diseases

- Sedimentology
- Hydrogeology
- Oceanography

NEED FOR PUBLIC AWARENESS

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Global Environmental Problems

- Over populations
- Pollution
- Depletion of Resources
- Global change
- War

Environmental Priorities in India

- Population Stabilization: Carrying capacity
- Natural resource assessment and management
- Strategies for environmental pollution monitoring and control
- Human settlement and public health
- Environmental Laws and regulation
- Environmental Education and Awareness
- Dynamic pressure of limited resources linking to Hazards and Disasters

- **"Tragedy of Commons"- individuals promote their short terms welfare, which conflicts with long-term environmental interest**

NEED FOR PUBLIC AWARENESS

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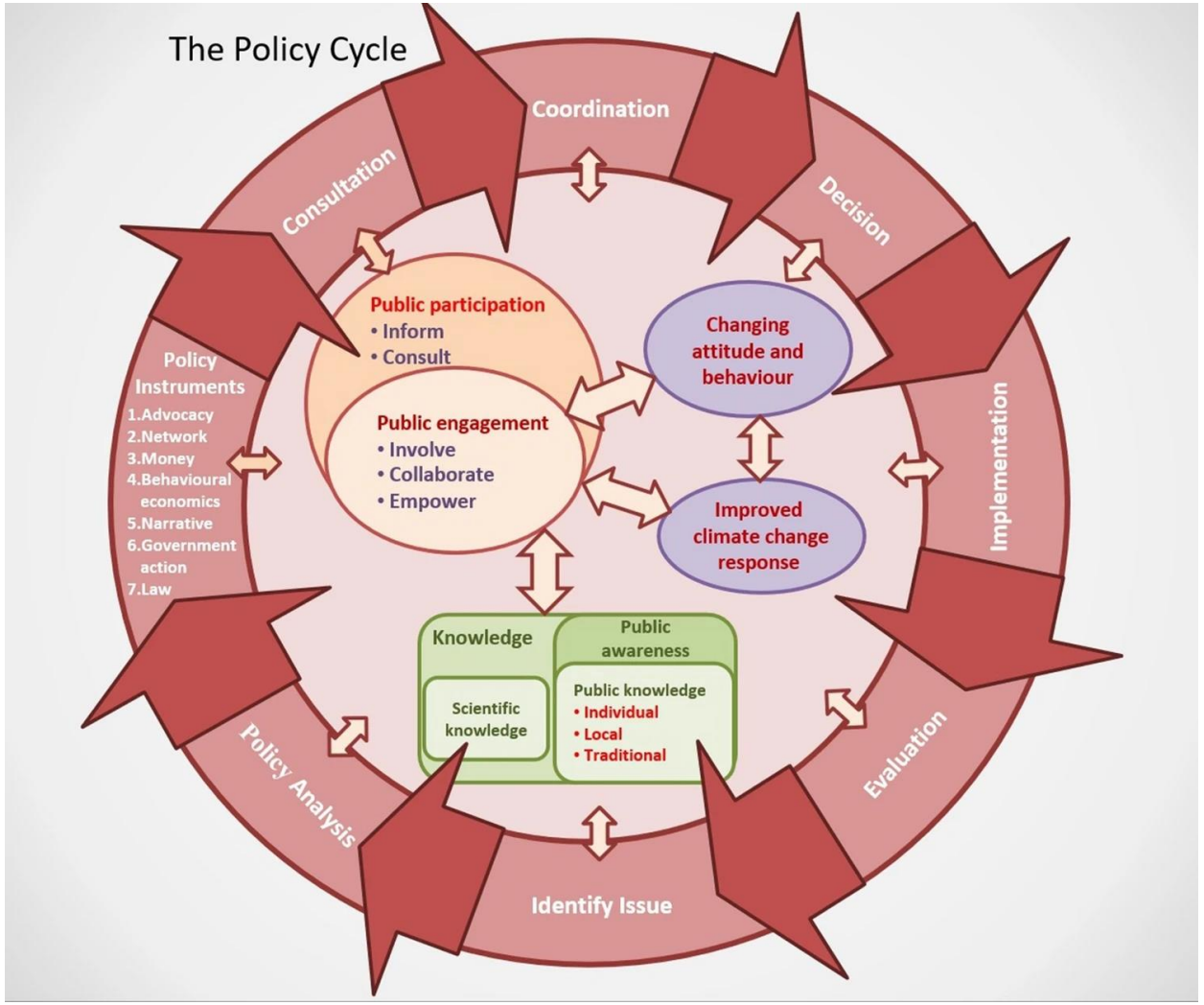
- Communities are vulnerable to the impacts of climate change
- Need to adapt to increase their resilience.
- The community needs to be made aware of the risks, acquire knowledge about the options that are available for a response, and be empowered to take their own actions
- Drivers for effective government policies



Khatibi, F.S., Dedekorkut-Howes, A., Howes, M. *et al.* Can public awareness, knowledge and engagement improve climate change adaptation policies?. *Discov Sustain* **2**, 18 (2021)

ECOSYSTEM

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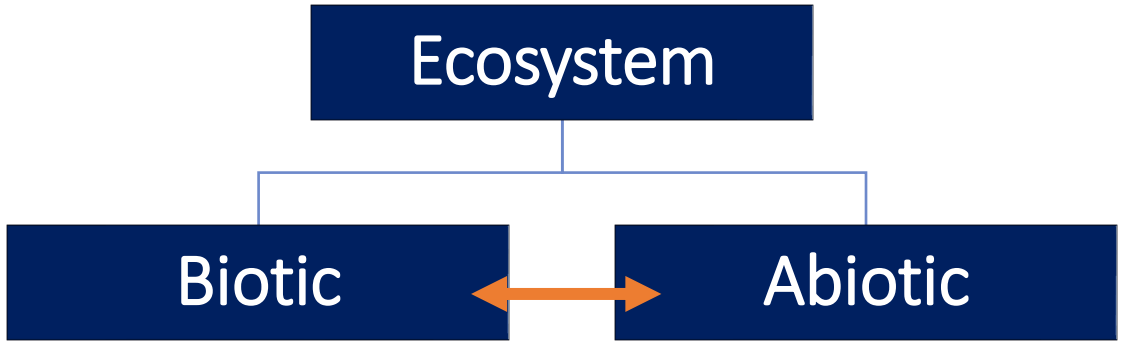
- interaction of a variety of individual organisms with each other (same or different species) and their physical environment.
- Self-contained system with specific nutrients and energy flow
- Individual→ Population→ Ecosystem → Biome→ Biosphere

- Ecological Principles



- **Producers**
- **Consumers**
- **Decomposers (Saprophytes)**
- **Scavengers**
- **Parasites**
- **Phagotrophs and Osmotrophs**

- **Climatic Conditions**
- **(Temperature, rainfall, sunlight, Wind, Atmospheric gases)**
- **Edaphic conditions**
- **(soil, pH, mineral nutrients)**

ECOLOGICAL PRINCIPLES

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- Homeostasis
  - Tendency of the body to seek and maintain balance within its internal environment even it is undergoing external change: RESILENCE
- Limiting Factor
  - Limiting factor determines the productivity of organism, population, or communities
- Law of thermodynamics
  - First law: Law of conversation of energy
  - Second Law: Disorder is constantly increasing, and during energy conversion, energy is converted into a less organized and useful form.
  - “Energy transfer from along the food chain in less than 10 %

MODELS FOR ENERGY FLOW IN THE ECOSYSTEM

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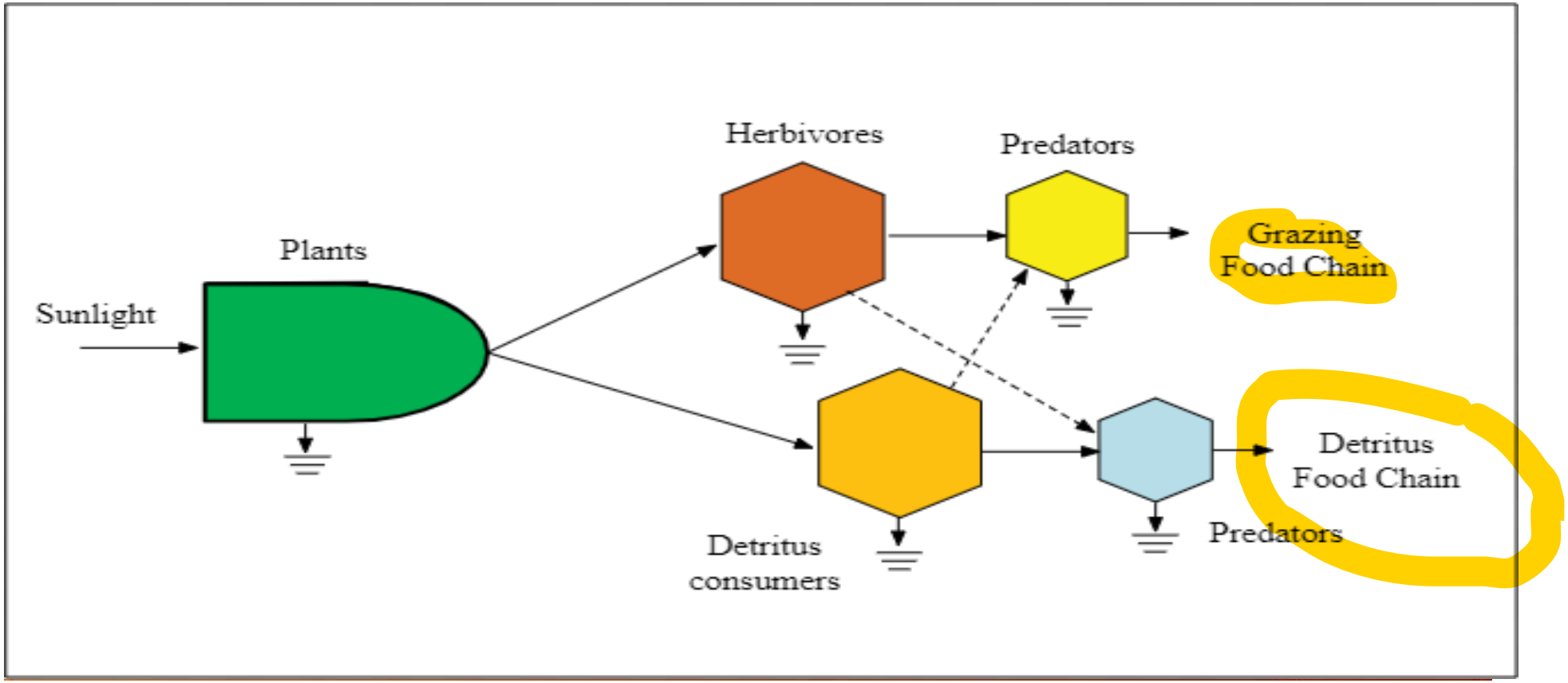
- Universal energy flow:
- Single Channel model
- Double channel or Y shape model



FOOD CHAIN AND FOOD WEB

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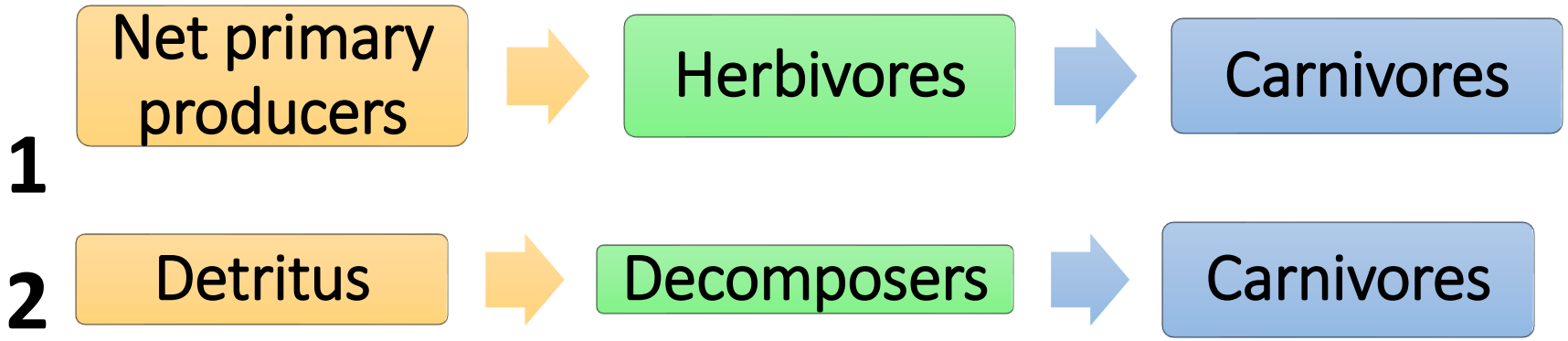
- Food Chain: unidirectional transfer of energy through the community
- Grazing food chain, Detrital food chain, Supplementary (Parasitic) food chain
- Detrital food chain is common to all ecosystem



- Food web is a graphical description of interconnected feeding relationships among species in a community.

PRODUCTIVITY

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- is the rate of formation of energy and matter, which is stored in different species at different trophic levels.
- Primary Productivity (dry wt. g calories/ unit area/ unit time): the rate at which chemical energy is produced from solar energy.
  - NPP of the earth is 320 dry g. calories / m2/ year

- Secondary Productivity: Rate with which energy is stored at different levels in consumers
- Gross Primary Productivity- is the total amount of organic matter produced through photosynthesis
- Net Primary Productivity- the amount of energy that remains available for plant growth

ECOLOGICAL PYRAMID

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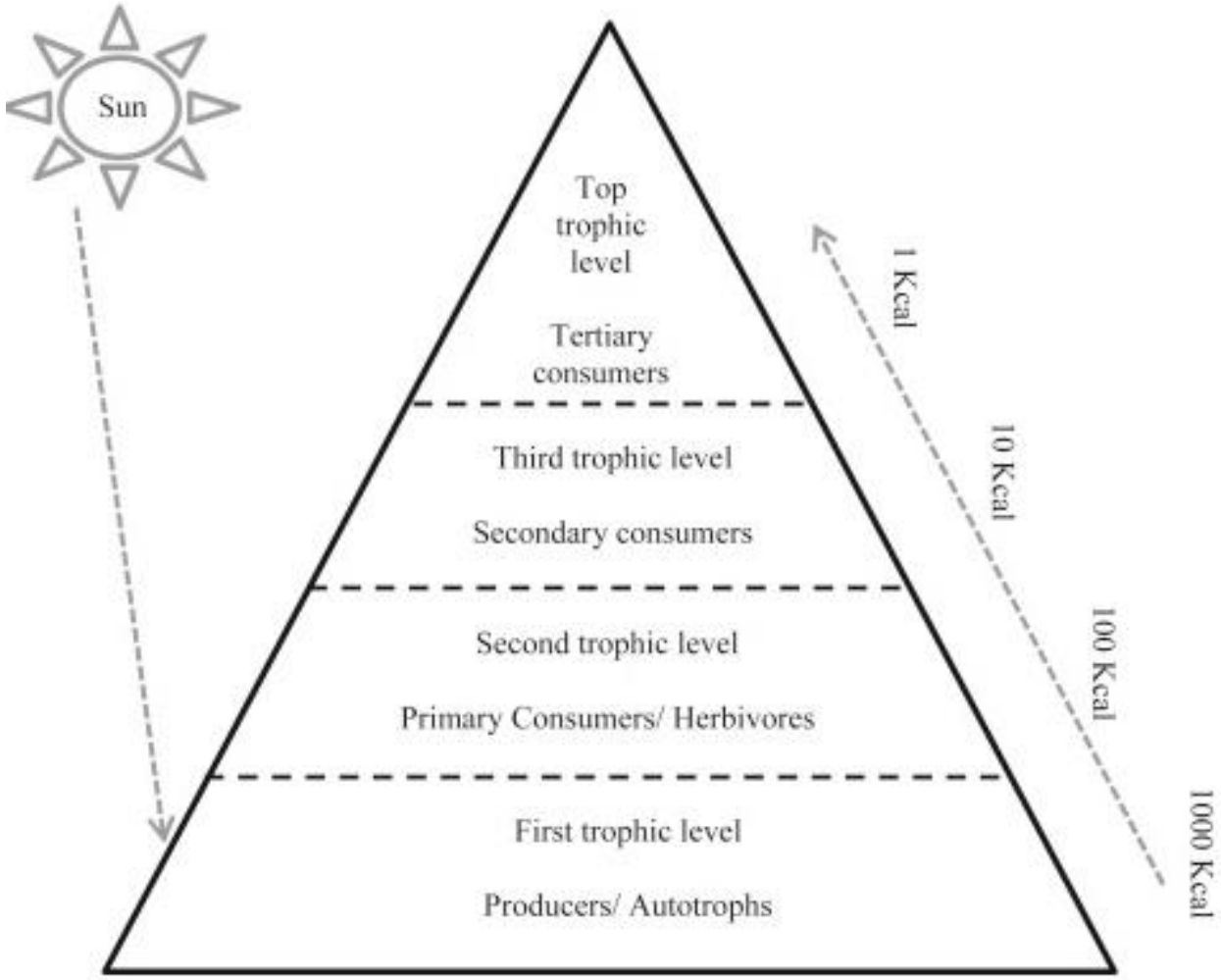
- **Eltonoian Pyramid is of three types:**

1. **Pyramid of energy**
2. **Pyramid of Number**
3. **Pyramid of biomass**

PYRAMID OF ENERGY

- **The pyramid of energy is always uptight**
- **Lindeman Law (10 % Law): When energy is transported from one trophic level to another, 90 % is used in the metabolic process, and only 10 % is transferred**



PYRAMID OF NUMBER

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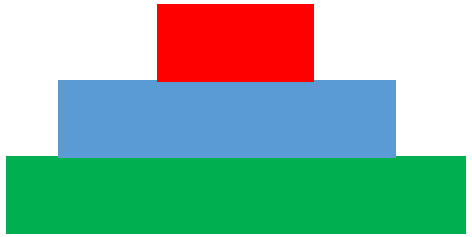
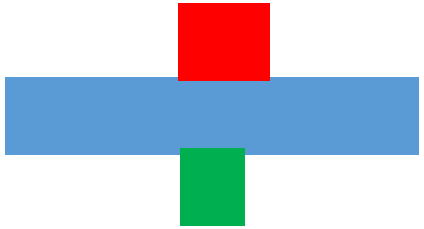
A pyramid of numbers indicates the number of individuals in each trophic level.



# of carnivores
# of herbivores
# of producers

Inverted pyramid: higher trophic level may have a larger standing crop than a lower trophic level.



bird
insects
tree

PYRAMID OF PRODUCTIVITY

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A pyramid of standing crop/ productivity indicates how much biomass is present in each trophic level at any one time.

# of carnivores

# of herbivores

# of producers

NUTRIENT CYCLING

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- Biogeochemical cycle
- Cycling of nutrients can be in terrestrial {nitrogen cycle} and aquatic {carbon cycle} .
- Cycling of nutrients can be:
  - Biological cycle.
  - Geological cycle

CARBON CYCLE

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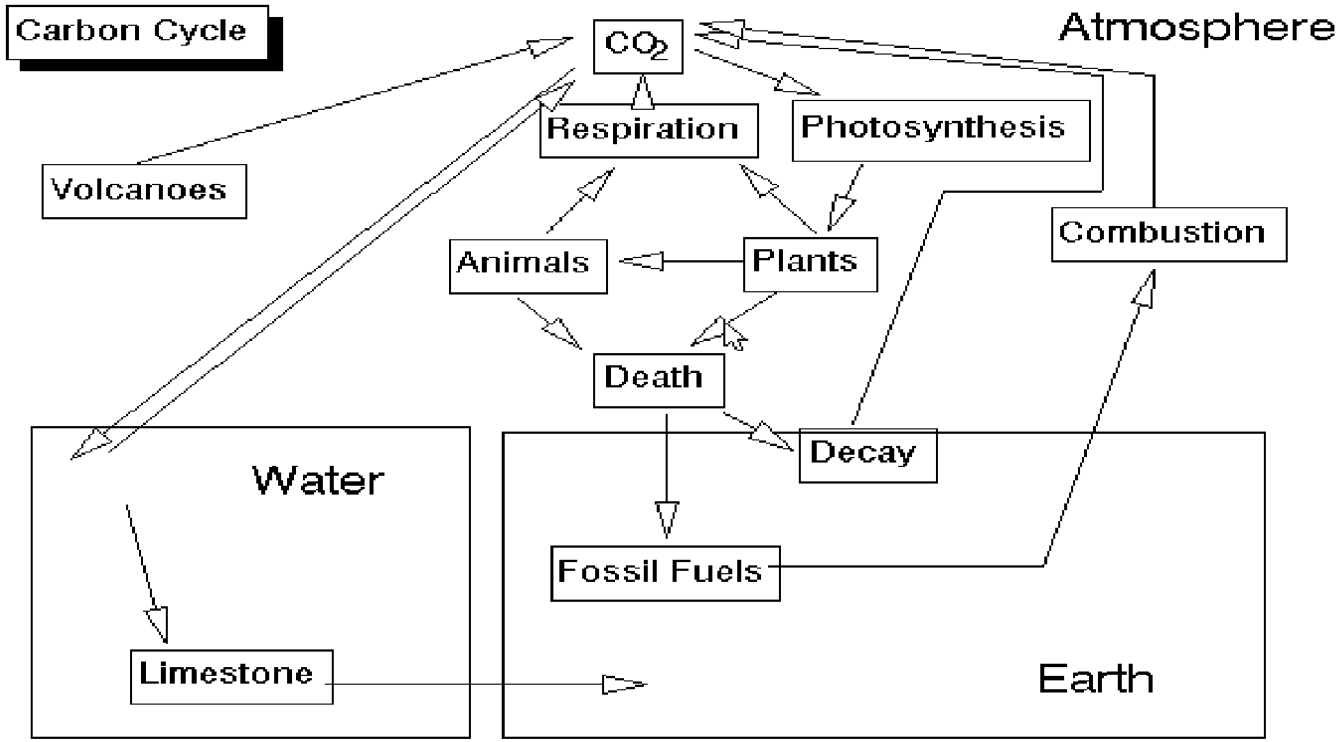
Reservoir:

- Organic {detritus materials, coal, natural gas} inorganic {air, sediments, oceans}

Fate:

- Removal of carbon dioxide from the air by photosynthesis, fossil fuels, water bodies (calcium bicarbonate)
- Return of carbon dioxide to the atmosphere by respiration, decomposition, combustion, volcanic activities, water bodies

- *Ocean is the major sink of carbon. Dissolved CO2 is also used by algae for production.*



Human impact :

- Introduce carbon into the atmosphere from longer-term stores. Increase the levels of carbon dioxide
- Greenhouse gas contributes to global climate change.
- Increased temperature decreases CO2 fixation in water and
- Ocean acidification

NITROGEN CYCLE

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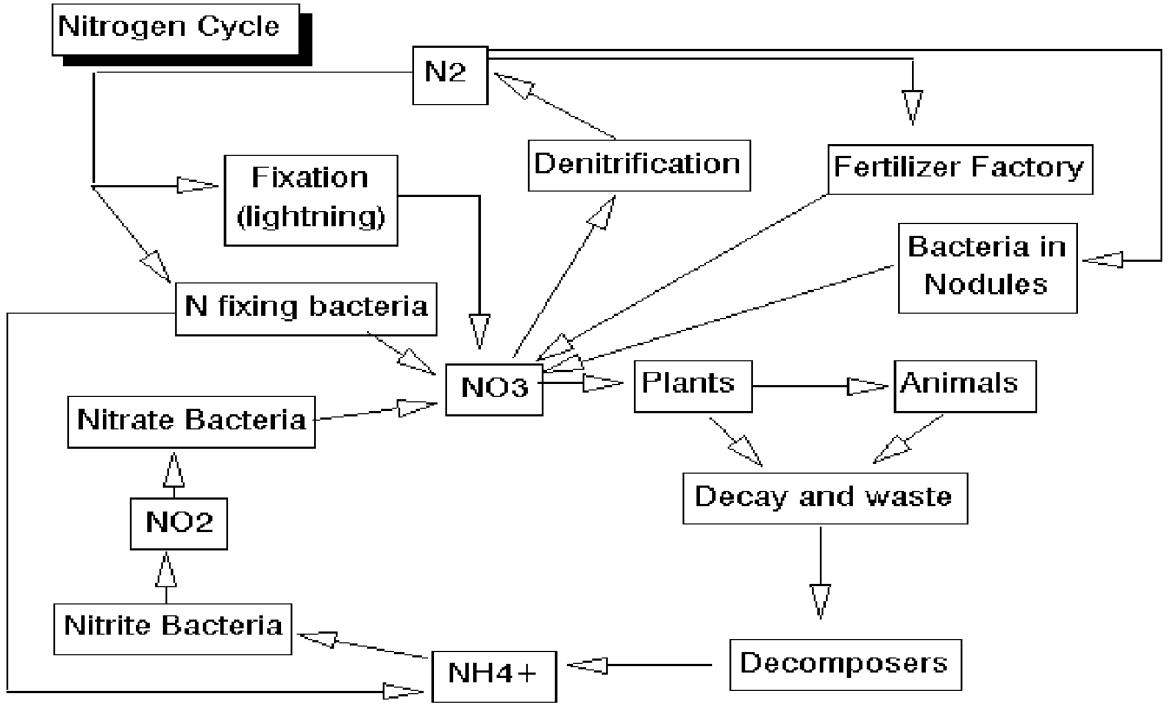
Reservoirs

Organic { biomass}, Inorganic (Air) nitrogen is 78% in the air but is unusable. The usable form is NH3 and NO3.



Fate

- Nitrogen (nitrate) is taken away from the soil by plants and chemosynthetic bacteria
- **Denitrifying bacteria** decompose nitrate in badly aerated soil e.g., water-logged soil
- Return of nitrogen to the soil by
  - **Ammonification:** decomposition by putrefying bacteria (anaerobic)
  - **Nitrification:** the ammonia or ammonium compound are oxidized (*Nitrosomonas*, *Nitrobacter)*
- **Lightning:** Nitrogen combines with oxygen which then dissolved in rain to form nitric acid
- **Industrial method: nitrogen combines with hydrogen to form ammonia**
- Direct fixation by free-living nitrogen-fixing bacteria *(Azotobacter), symbiotic nitrogen-fixing bacteria (Rhizobium)*

Human impact

- Acid rain
- Eutrophication

PHOSPHOROUS CYCLE

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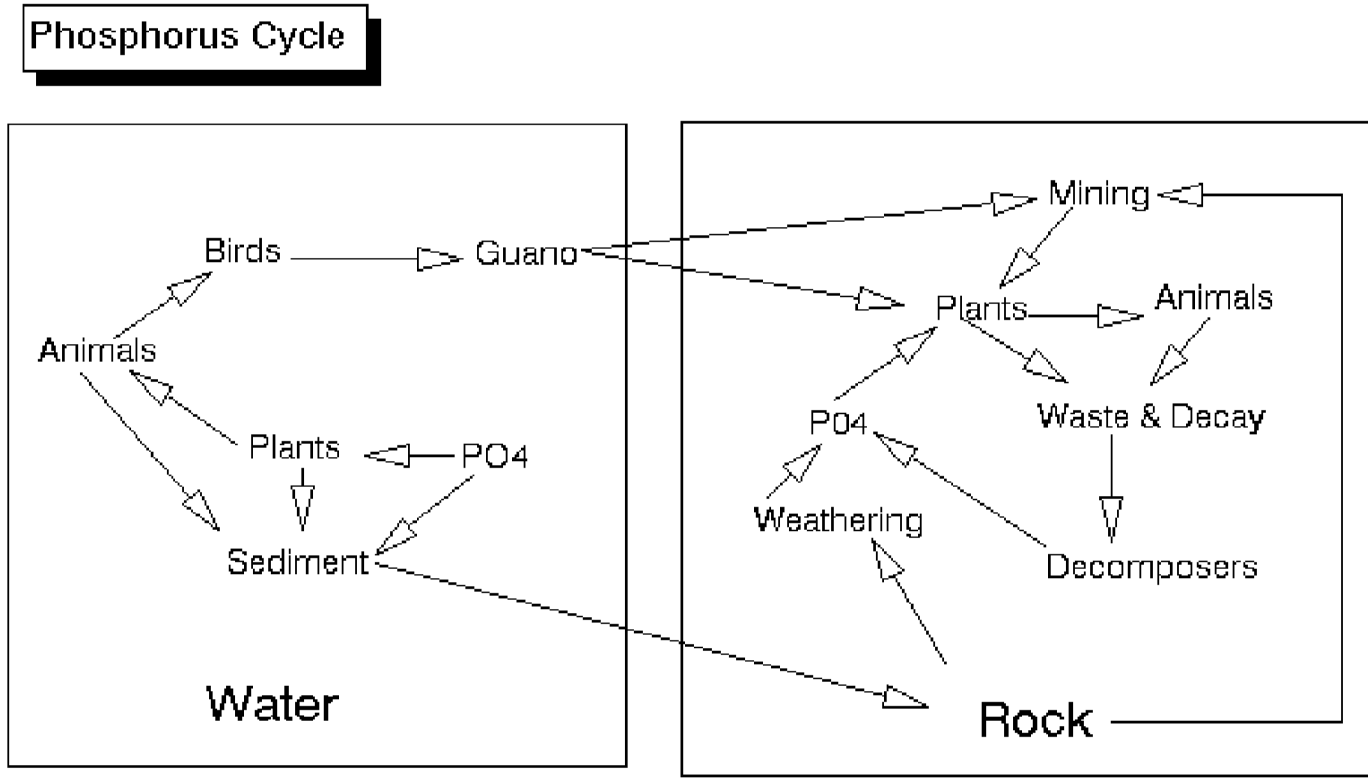
Reservoir:

- Rocks and soil minerals, oceanic sedimer
- geologic uplift or mountain building

$$P4\ (s)+502\ (g) \quad P_4O_{10}(s)$$

Human impact

- Commercial fertilizers and phosphate-containing: Eutrophication
- Slash-and-burn forest clearance reduces phosphate levels, Mining activities.
- Washing of phosphate in land



SULFUR CYCLE

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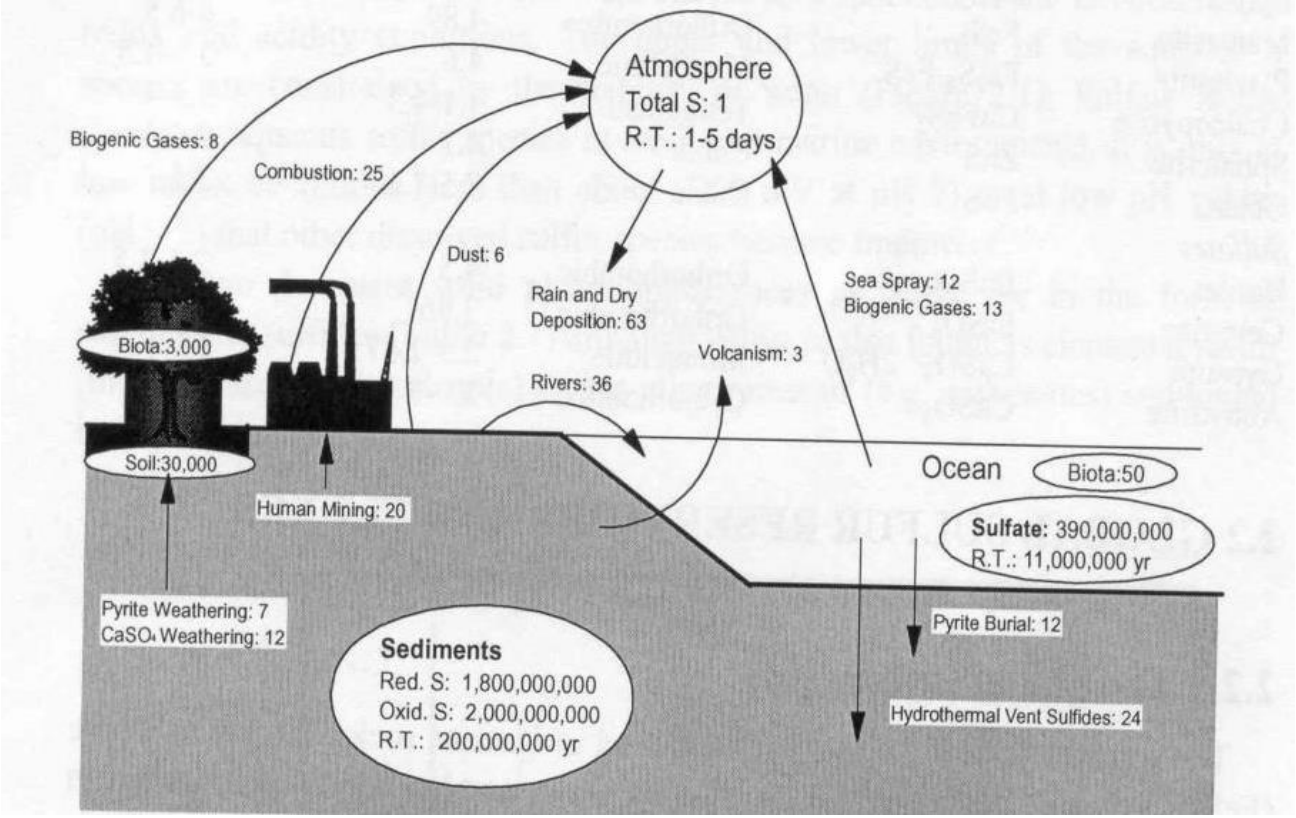
Reservoirs

Rocks and minerals, including as sulfate salts buried deep within ocean sediments.

Fate:

- The sulfur cycle contains both atmospheric and terrestrial processes.
- Weathering of rocks, releasing the stored sulfur.
- Oxidation of sulfur into sulfate (SO4).
- Uptake of sulfate
- Emission of sulfur directly into the atmosphere, including volcanic eruptions,
- Breakdown of organic matter in swamps and tidal flats



TYPES OF ECOSYSTEM

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Ecosystem

- Terrestrial
- Aquatic

- *Natural*
- *Artificial*

- Forest ecosystem
- Deserts ecosystem
- Grassland ecosystem

- Ponds (lentic)
- River (lotic)
- Lakes (lentic)
- Oceans
- Deep water
- Estuaries
- Coral reefs
- Wetlands/ Salt Marsh

- Freshwater
- Marine

Aquatic Ecosystem

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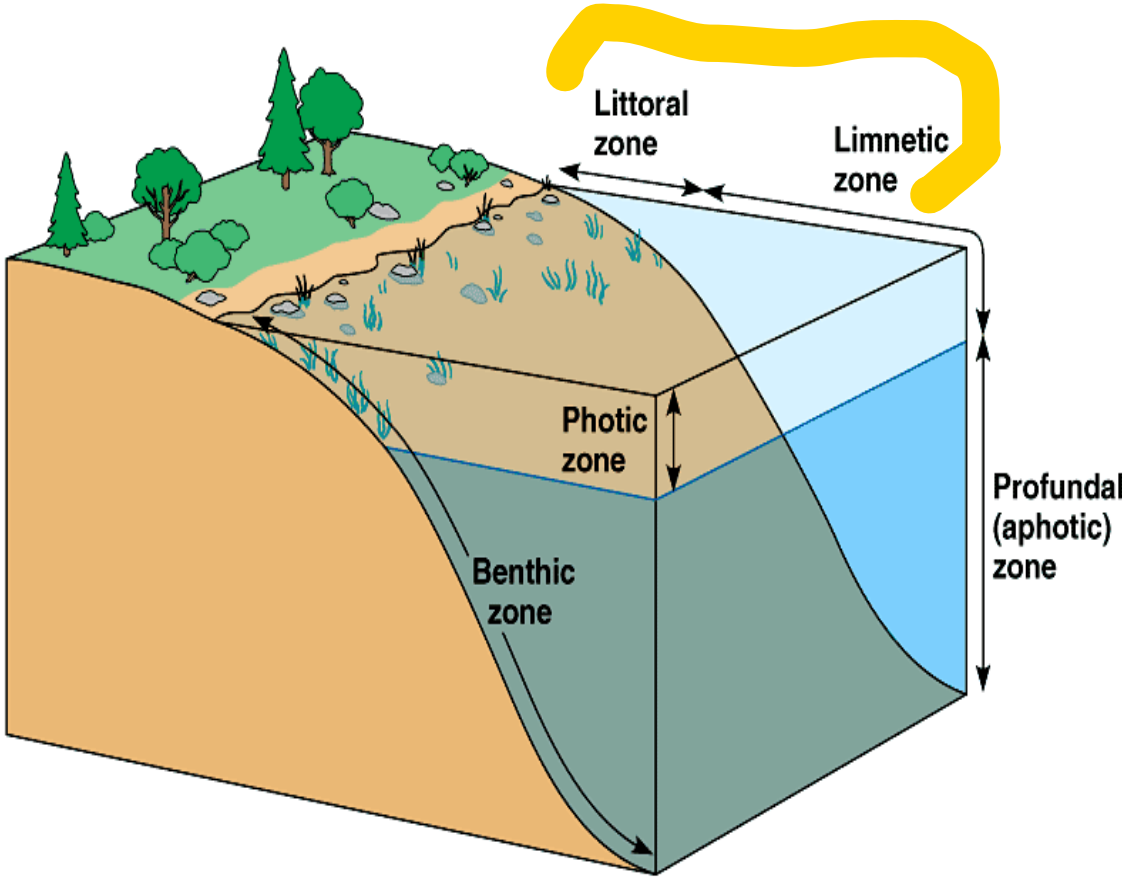
Lentic/ Lotic

- Based on solar irradiation
  - **Euphotic Zone:** Presence of Sufficient light for photosynthesis
    - **Littoral:** a shallow zone near the shore. Rooted plants, phytoplankton
    - **Limnetic Zone:** well-lit zone in open sea, running water, Species found are Zooplanktons and nektons
  - **Profundal zone:** lack of sufficient sunlight and oxygen. Mostly occupied by consumers
  - **Benthic Zone:** No sunlight and oxygen.



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Aquatic Ecosystem

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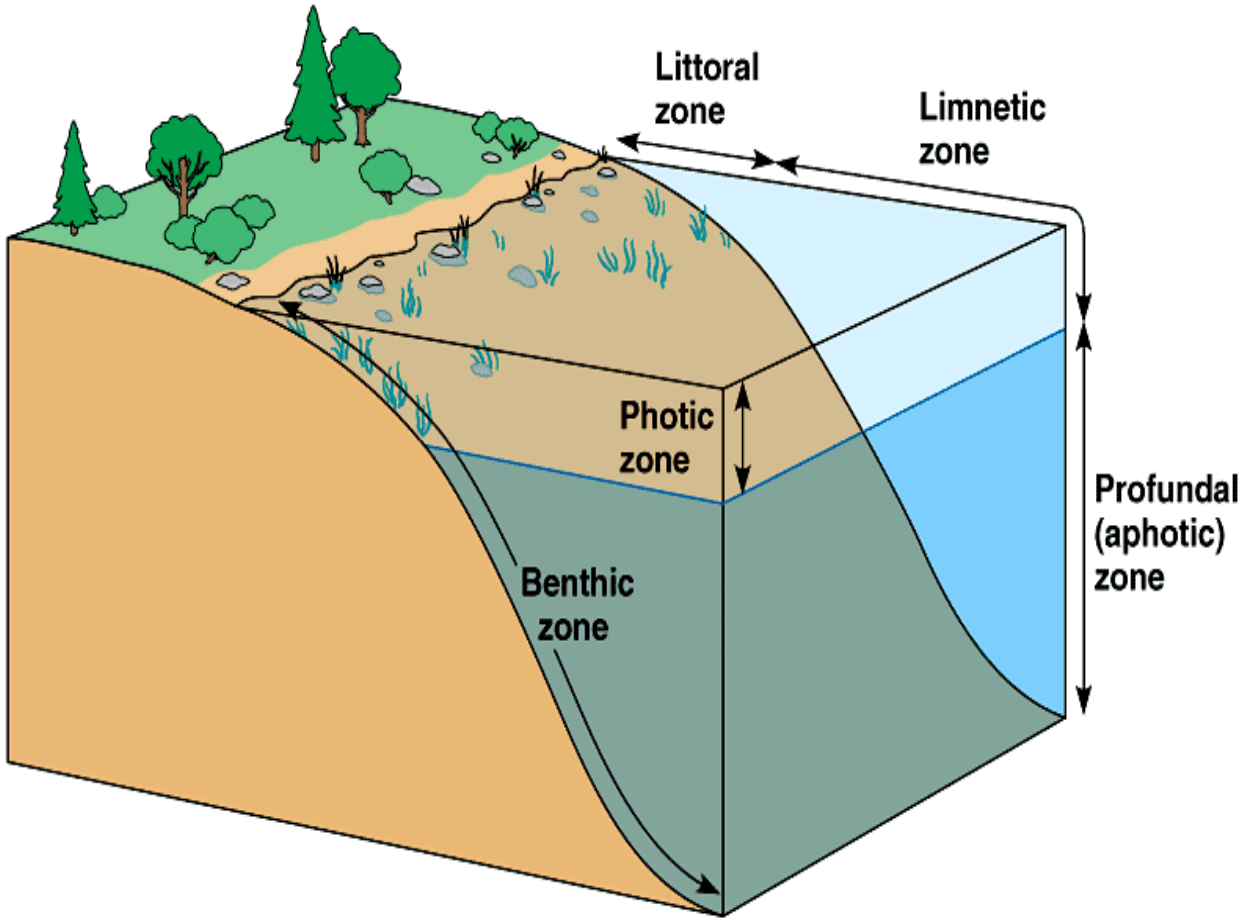
Lentic/ Lotic

- Based on productivity/ nutrient content
  - **Oligotrophic:** less productive lake. Usually deep and step side with narrow littoral zone
  - **Eutrophic:** High productive lake, rich in organic matter and nutrient
  - **Mesotrophic:** intermediate level of productivity, medium-level nutrients, usually clear water with submerged aquatic plants.



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Marine Ecosystem

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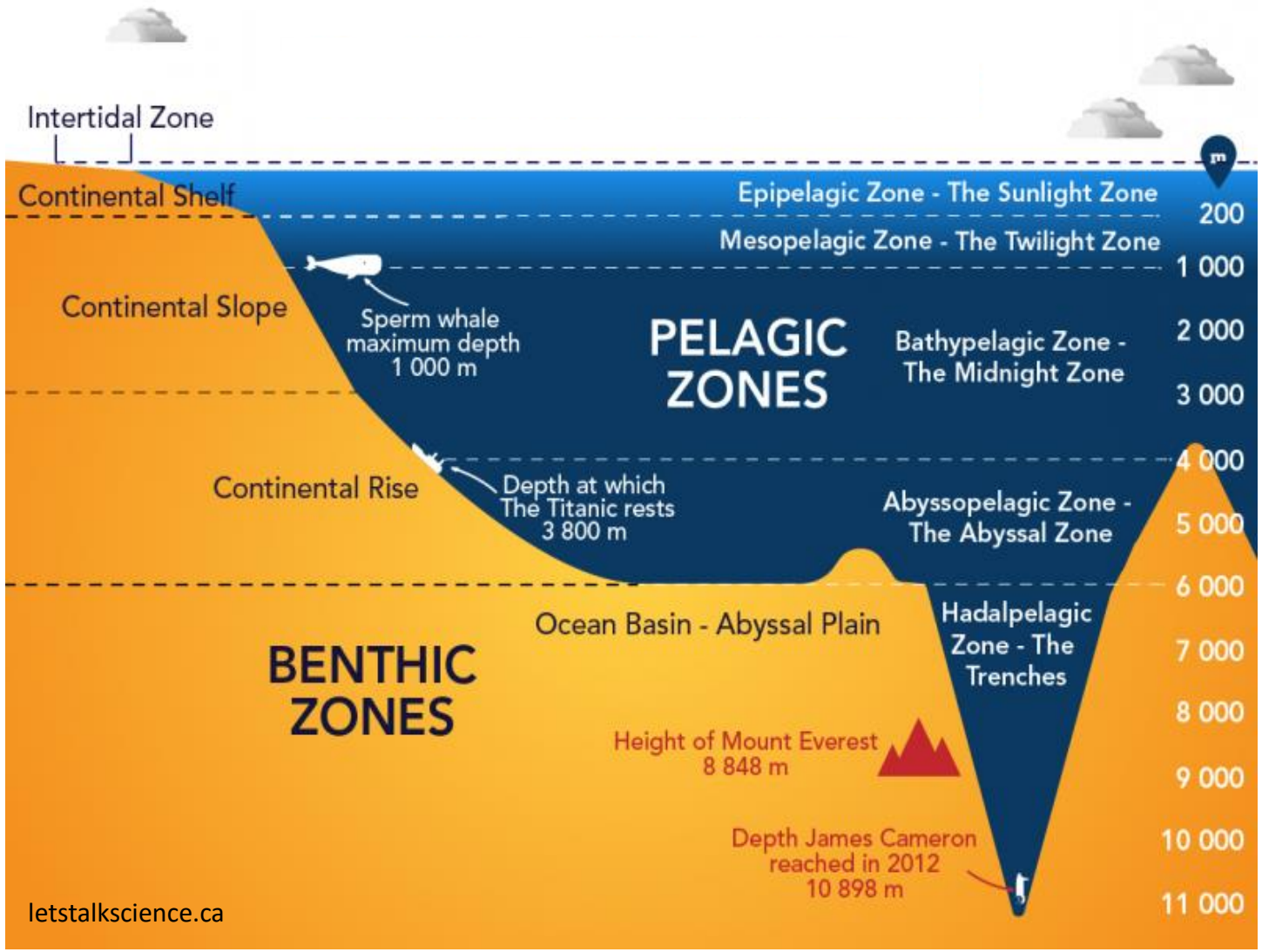
- Oceans
- Deep water
- Estuaries
- Coral reefs
- Wetlands/ Salt Marsh

- Pelagic and Benthic

  Benthic: Supralittoral, littoral, Sublittoral, Bathyal, Abyssal, Hadal, trenches

  Pelagic: Neritic (high biodiversity, nutrient), Oceanic



Marine Ecosystem

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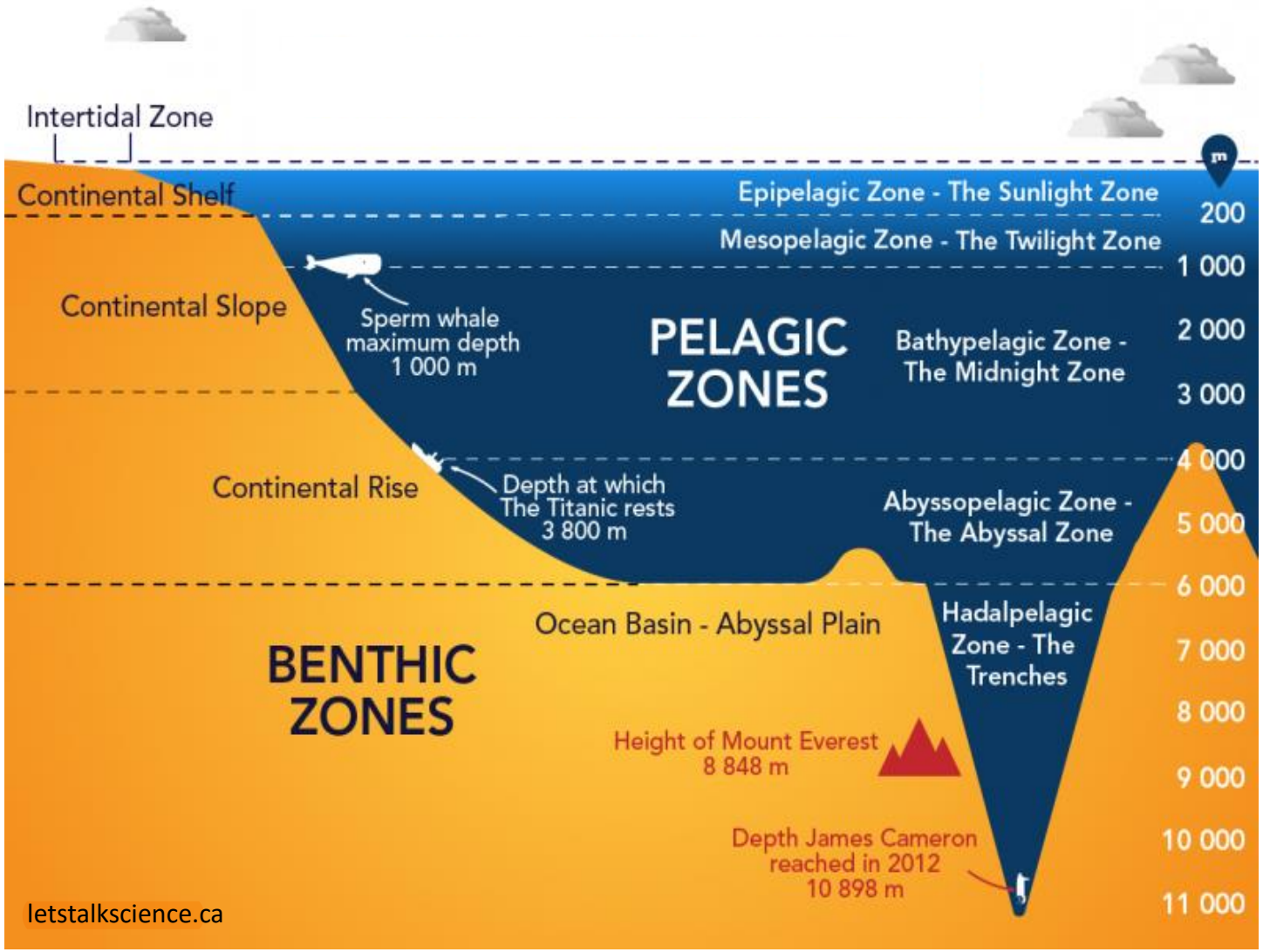
- Oceanic based on the light
  - Photic,
  - Euphotic,
  - Disphotic
  - Aphotic
- Oceanic based on depth
  - Epipelagic,
  - Mesopelagic,
  - Bathypelagic
  - Abyssopelagic



Wetlands

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- "A wetland can be defined as an area of marsh, fen, peatland or water, natural or artificial, permanent or temporary with water static, flowing, fresh, brackish, salt, including area of marine water and depth less than 6m at the time of low tide (Ramsar Convention)

Function:

1. Biodiversity,
2. Groundwater replenishment, Water filtration, S
3. Sediment retention, Erosion control,
4. Tourism, Retention of nutrients,
5. Mitigation against climate change,
6. Natural transport infrastructure

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Types

- Marshes: shallow water that is mostly grassland. Can be fresh and saltwater
- Swaps: slow-moving streams, rivers, deeper than marshes. Have wood shrubs rather than grass
- Bog and Fens: freshwater wetlands formed by glacial lakes. Low nutrients
- Estuaries: Enclosed bodies where freshwater meets salt water, high biodiversity, with brackish water

Coral reef

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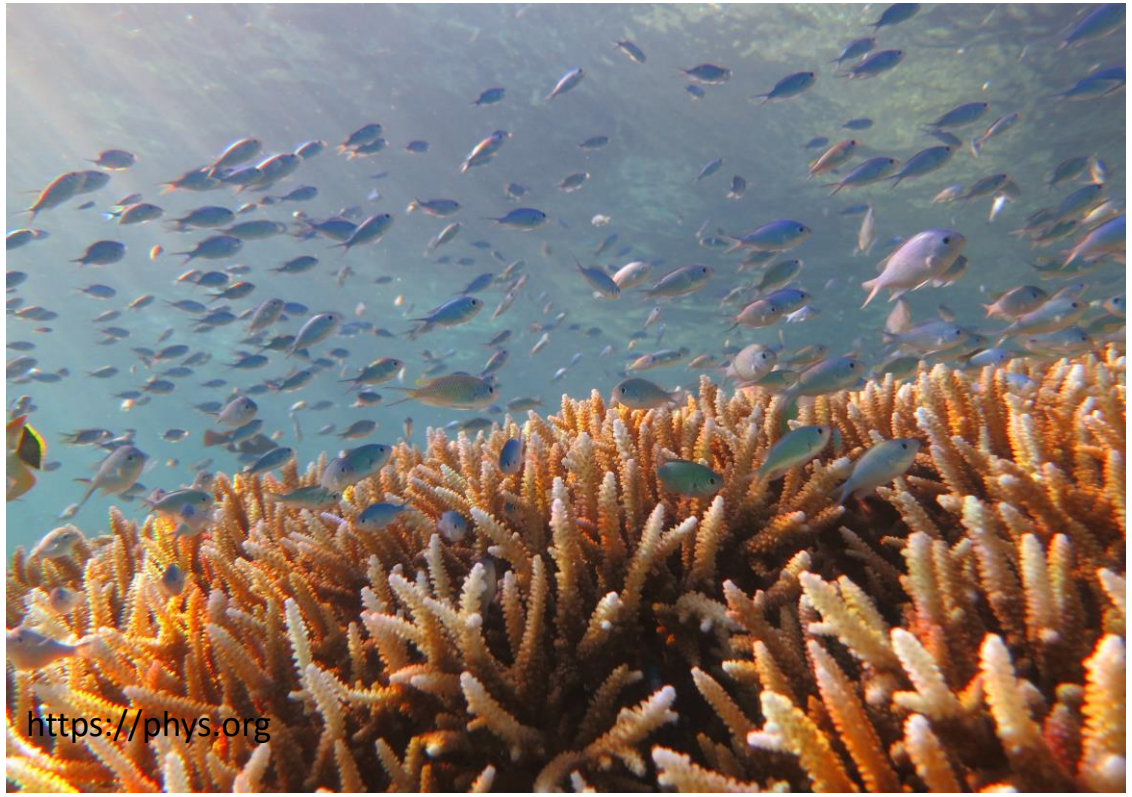
- diverse underwater (0- 60 m) ecosystem held by calcium carbonate secreted by corals.
- Individual coral is called a polyp,
- Shallow water corals feed on Zooxanthellae (algae)-
- Zooxanthellae provide colors to corals
- Temp range for survival- 20-21 degree C, Salinity 27-30 %



- Types
- Fringing Reefs: They are attached to the margins of continents. They form in areas where the rainfall-runoff is relatively low.
- Barrier Reefs: A lagoon separates them from land. They either form around islands or as lines parallel to the continental shore.
- Atolls: It is a ring-shaped island made up of coral reefs along with coral debris. It surrounds a shallow lagoon, completely or partially from which no land projects

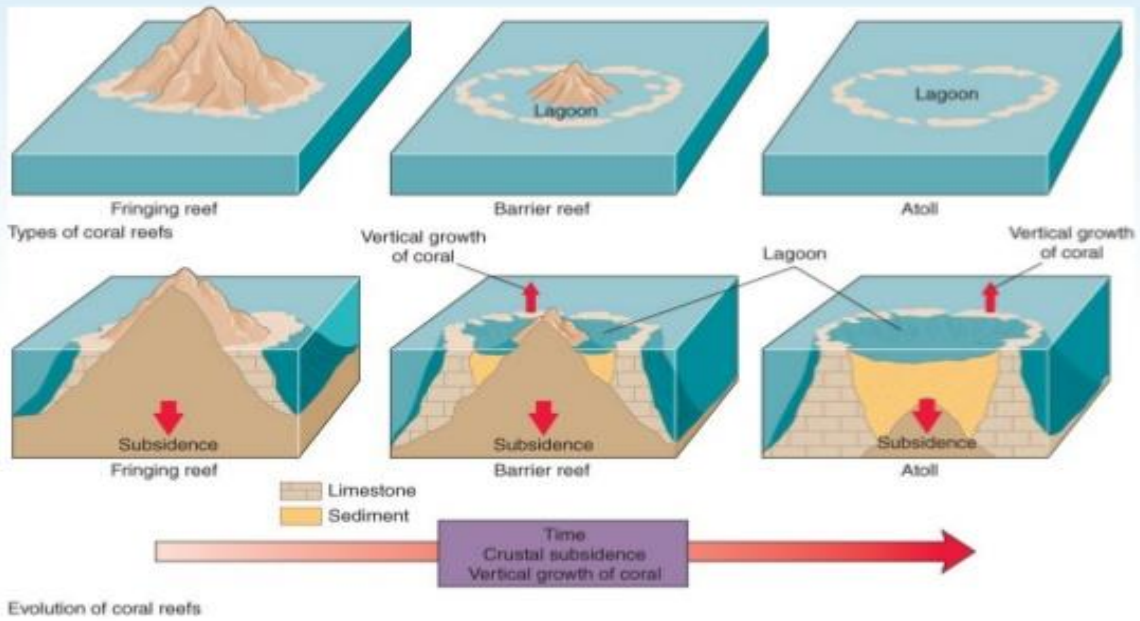


Fig. 9.8 The developmental sequence of coral reefs, from young fringing reefs (left), to barrier reefs (center), and finally to atolls (right).

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Terrestrial Ecosystem

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Forest

1. Tropical Rain forest
2. Tropical Deciduous forest
3. Tropical scrub forest
4. Temperate Rain forest
5. Temperate Deciduous forest
6. Evergreen coniferous/ Boreal

Grassland

7. Tropical: Savannas
8. Temperate forest
9. Polar / Arctic Tundra

Desert

10. Tropical
11. Temperate
12. Cold

A. *Abiotic component: organic and inorganic*
B. *Biotic component:*
   i. *Producers:*
   ii. *Primary, Secondary, and Tertiary consumers*
   iii. *Decomposers*
C. *Services rendered: Ecological and Commerical*

ECOLOGICAL SUCCESSION

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- Directional and sequential change of plant or community or ecosystem over time in orderly sequence and later stabilised to form climax
- Hydrarch
- Mesarch
- Xerarch
- Process of Succession:
  - Nudation
  - Invasion
  - Competition
  - Reaction
  - Stabilisation
- Species: Pioneer Middle and Climax

- Ecotone (transition between two communities) and Ecocline (joint expression to the communities)
- Ecological niche web of relationship of member of flora and fauna in an ecosystem/ community.

BIODIVERSITY

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- is defined as the variability among the living organism from all ecosystems.
- Describe how much variety an ecosystem has.

Level of Biodiversity

- Species: refers to the variety of species within a region
- Genetic: refers to the genetic variability within species. Each individual species possesses genes that are the source of its own unique feature.
- Ecosystem: variety in an ecosystem in a given place

- Alpha: refers to organisms interacting and competing for the same resources or sharing the environment
- Beta: refers to an expression of diversity between habitats
- Gamma: diversity of habitat within a landscape

VALUE OF BIODIVERSITY

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Value categories		Level of operation
Commodity value	Utilization value of organisms to humans as food, medicine	Population and species diversity
Amenity value	The value of human visiting and viewing or learning	Species and ecosystem diversity
Ecological integrity value	the importance of maintaining all functionally critical aspects of the ecosystem	Functional diversity
Ethical value	Moral obligation on human not to drive other species to extinction	Species and ecosystem diversity
Option value	The future evolutionary potential of species which can latter be used or commodity value	Genetic and population diversity

BIODIVERSITY HOTSPOTS

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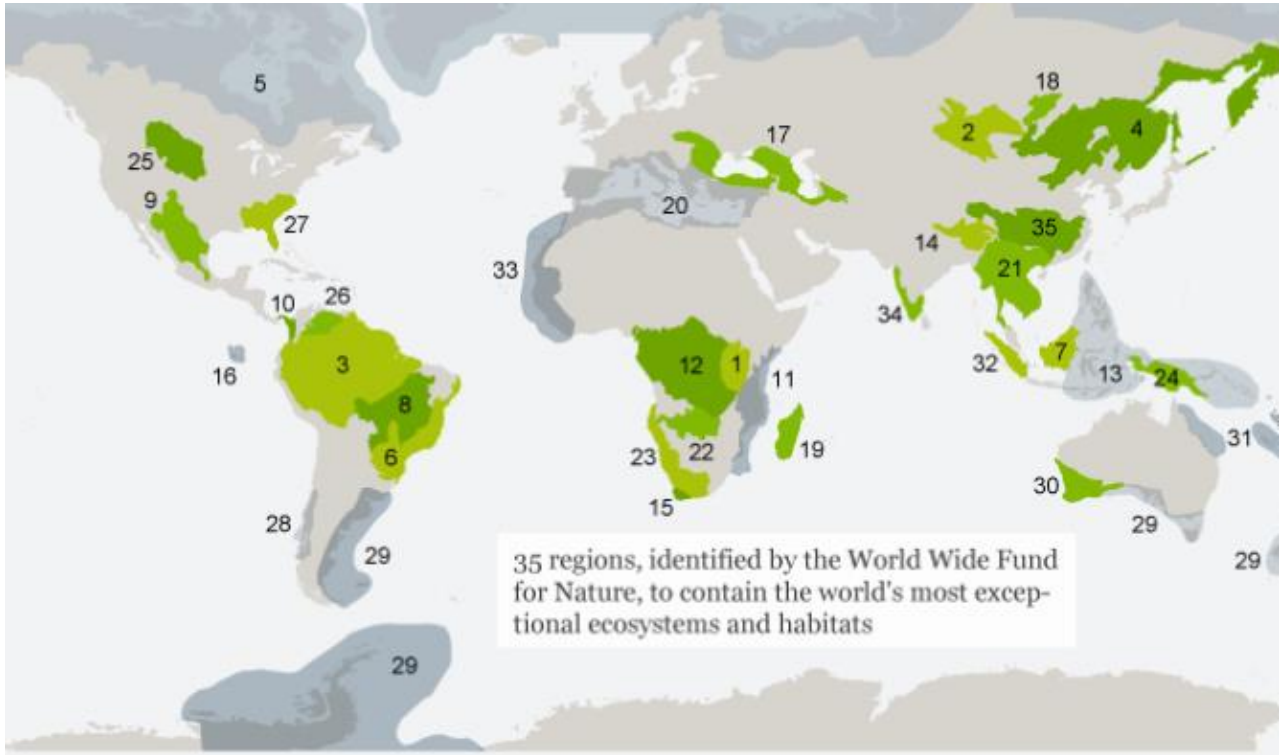
- Are areas that support natural ecosystems that are largely intact & where native species and communities associated with this ecosystem are well represented
- Endemic Species
- at least 1,500 vascular plants as endemics —a high percentage of plant life found nowhere else on the planet.
- A hotspot, in other words, is irreplaceable.
- It must have 30% or less of its original natural vegetation. It must be threatened.



1: African Rift Lakes Region **2**: Altai-Sayan Montane Forests **3**: Amazon Guianas **4**: Amur-Heilong **5**: Arctic Seas **6**: Atlantic Forests **7**: Borneo **8**: Cerrado-Pantanal **9**: Chihuahuan Deserts **10**: Choco-Darien **11**: Coastal East Africa **12**: Congo Basin **13**: Coral Triangle **14**: Eastern Himalayas **15**: Fynbos **16**: Galapagos **17**: Greater Black Sea Basin **18**: Lake Baikal **19**: Madagascar **20**: Mediterranean **21**: Mekong Complex **22**: Miombo Woodlands **23**: Namib-Karoo-Kaokoveld **24**: New Guinea & Offshore Islands **25**: Northern Great Plains **26**: Orinoco River & Flooded Forests **27**: South-eastern Rivers & Streams **28**: Southern Chile **29**: Southern oceans areas **30**: South-west Australia **31**: South-west Pacific **32**: Sumatra **33**: West Africa Marine **34**: Western Ghats **35**: Yangtze Basin

Source: WWF | Wildlife in a Warming World, 2018

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THREAT TO BIODIVERSITY

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- Loss of species, ecosystem, genes, and extinction
- Can cause:
  - Affects restoration and resilience
  - Human-caused stress: Climate change
  - Natural disaster
- Reasons:
  - Habitat destruction
  - Poaching

BIODIVERSITY CONSERVATION METHODS

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- CONSERVATION OF BIODIVERSITY:
  - Objective:
    - Maintain essential ecological processes
    - Preserve diversity of species
    - Make sustainable utilization of species and ecosystem,

In-situ

- National Park, Biosphere Reserves, Wildlife Sanctuaries, Wetlands
- Biodiversity Hotspots
- Scared Trees and Lakes
- Natural Gene pool

Ex-situ

- Gene Bank
- Seed Bank
- Botanical garden
- Zoological Garden

BIODIVERSITY CONSERVATION METHODS

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In-situ

- OBJECTIVE

1. Create protected areas
2. Promote sustainable management
3. Protect and restore of degraded land
4. Integrate of conversation and biodiversity
5. Create natural corridors

Ex-situ

- OBJECTIVE

1. Relocate endangered species from their natural habitat corridors

BIODIVERSITY PROTECTED AREA

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- National Parks: ecosystem where flora, fauna, geomorphic sites, and habitat are studied in the scientific, educational, and natural landscapes. It is in use for conservation purposes which reserve a natural, semi-natural, or developed land of the state.
- Wildlife sanctuary: declared for the purpose of protecting, propagating, and developing wildlife and the environment. Certain rights are allowed.

BIODIVERSITY PROTECTED AREA

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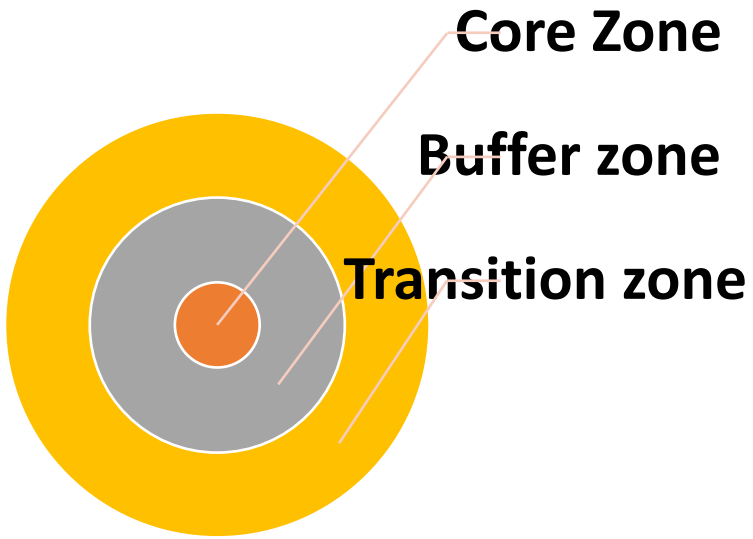
- Biosphere reserves: terrestrial and coastal ecosystem promoting solution to reconcile biodiversity conservation with sustainable use. It is nominated by the Central govt and remains within the jurisdiction of the State.



- Man and Biosphere Program 1971

BIODIVERSITY PROTECTED AREA

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Eco-Sensitive Zones (ESZs): The National Wildlife Action Plan (2002-2016) stipulated that state governments should declare land falling **within 10 km of the boundaries of national parks and wildlife sanctuaries as eco-fragile zones or Eco-Sensitive Zones (ESZs)**

What are the Activities Allowed in ESZs?

- **Prohibited Activities:** Commercial mining, industrial, hydroelectric projects (HEP), and commercial use of wood, Tourism activities.
- **Regulated Activities**: Establishment of hotels, change of agriculture system, widening of roads.
- **Permitted Activities:** Ongoing agricultural or horticultural practices, rainwater harvesting

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What is the Significance of ESZs?

- Minimize the Impact of Development Activities:
- In-situ Conservation:
- Minimize the Negative Impact on Fragile Ecosystems: 'Shock Absorber'

INTERNATIONAL CONVENTION FOR BIODIVERSITY CONSERVATION

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- International Union for Conservation of Nature, 1948: valuing and conserving nature, ensuring effective and equitable governance of its use, and deploying nature-based solutions to the global challenge.

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- Red Data List, 1964
  i. Extinct: the last individual has died or is nontrackable: Ex. Indian Cheetah
  ii. Extinct in the Wild: Species survive only in the captivity
  iii. Critically Endangered: Species that are at high risk of extinction; population declined by 80-90 % over the decade
  iv. Endangered: Very high risk of extinction as the population has declined by 50 -70% in a decade. Current population size (250)
  v. Vulnerable: High risk of endangerment in the wild
  vi. Near Threatened: Close to becoming threatened or may meet the criteria for threatened status.
  vii. Least Concern
  viii. Data Deficient
  ix. Not Evaluated

INTERNATIONAL CONVENTION FOR BIODIVERSITY CONSERVATION

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- Convention on International Trade in Endangered Species in Wild Fauna and Flora (CITIES)
  - Aim to ensure international trade in the specimen doesn't threaten their survival
  - Reduce economic incentive to poach endangered species and destroy their habitat
- Convention on Convention of Migratory Species of Wild Animals (CMS)
- Green Pease: opposing environmental abuse, saving endangered species, and increasing public awareness
- United nation Environment Program
- World Wide Fund for Nature
- TRAFFIC: The wildlife trade monitoring network is a strategic alliance between IUCN and WWF to ensure trade of wild plants and animals is not a threat to nature.

NATIONAL CONVENTION, ORGANISATION, & PROJECT OR BIODIVERSITY CONSERVATION

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- Wildlife Protection Act, 1972
- National Tiger Conservation Act
- The Biodiversity Act, 20222
- Zoological Survey of India
- Botanical Survey of India
- Gene Campaign

- Madras Crocodile Bank tryst (MCBT)
- Salim Ali Centre of Ornithology and natural History
- Wildlife Institute of India

- Project Tiger
- Project Snow Leopard
- Project Elephant
- SAVE (Saving Asia' s Vulture from Extinction
- Rhinoceros Conservation
- Ganges River Dolphin Conservation